

Exploring a desktop computer



For Internal Circulations Only

Schedule of Unit I

Class Schedule:

- Course overview, Exploring a desktop computer (Week 1, Class 1)
- History of computing (Week 1, Class 2)
- Exploring computing devices: Smartphones, Embedded systems in appliances, etc. (Week 2, Class 1)
- Quiz 1, Simulating Arduino circuits in Tinkercad (Week 2, Class 2)
- Project work (Week 3, Both sessions)

By the end of this session

- Recognize salient features of computing devices and systems and describe them.
- Identify external and internal hardware and software components of a desktop computer.
- Describe the booting process and role of the operating system.
- Perform basic desktop operations (file handling, checking system info).

What Is a Desktop Computer?

What Is a Desktop Computer?

- The word computer comes from the word “compute”, which means, “to calculate”
- Thereby, a computer is an electronic device that can perform arithmetic operations at high speed

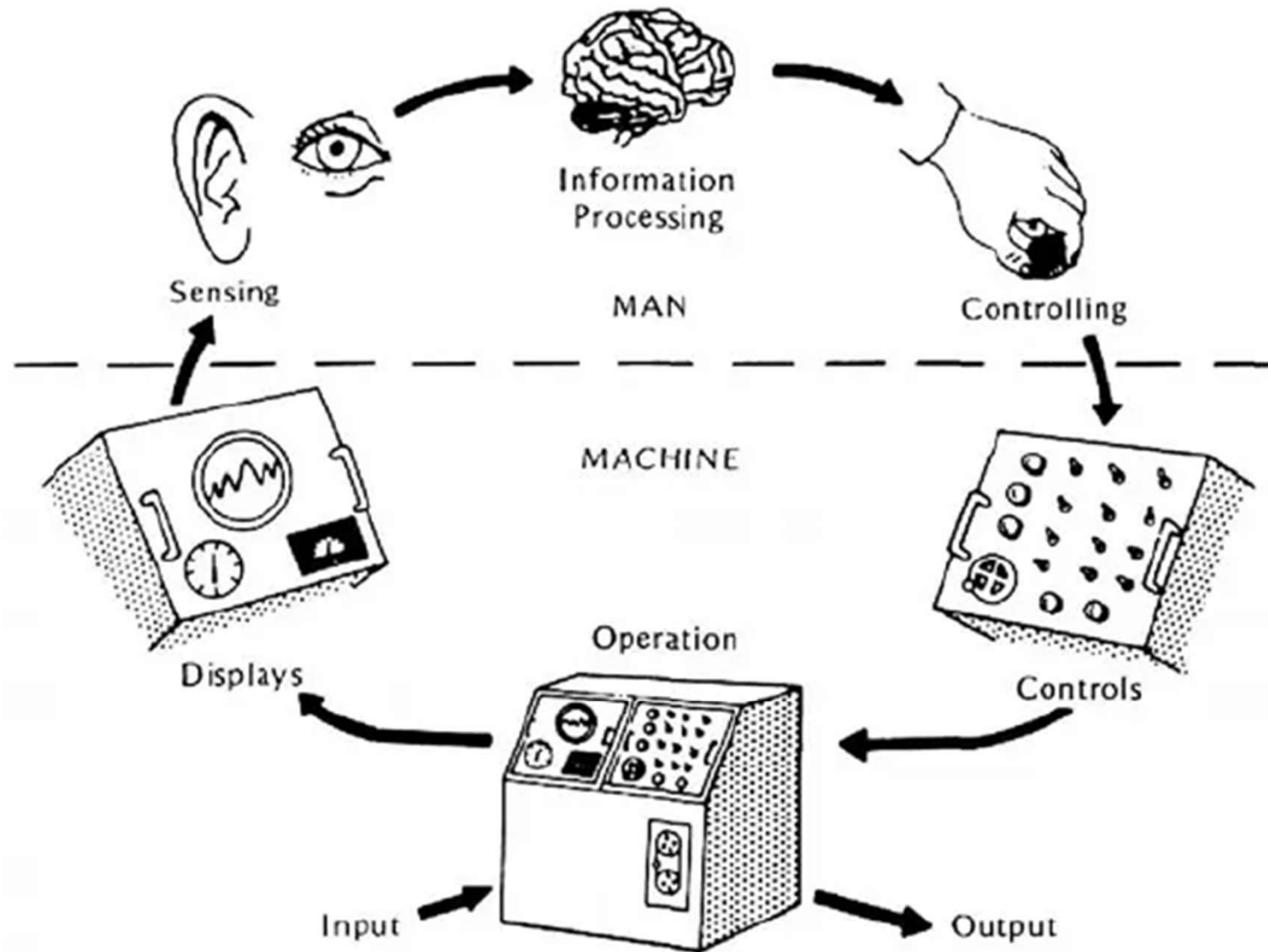
What Is a Desktop Computer?

- A **desktop computer** is a personal computer designed for **regular use at a single location**—typically on or under a desk—rather than being portable like a laptop or tablet.
- It is composed of multiple separate parts, including a central processing unit (CPU), monitor, keyboard, and mouse.

What Is a Desktop Computer?

Feature	Desktop	Laptop	Tablet
Portability	No	Yes	High
Performance	High	Medium–High	Limited
Upgradeability	Easy	Limited	Not possible
Screen Size	Large	Medium	Small

What Is a Desktop Computer?



Human machine interaction loop (Illustration: A Chapanis)

External Components



External Components - Ports

Thunderbolt 2



HDMI



Display Port



Lighting



USB B 3.0



USB Type A



Thunderbolt 4



Ethernet



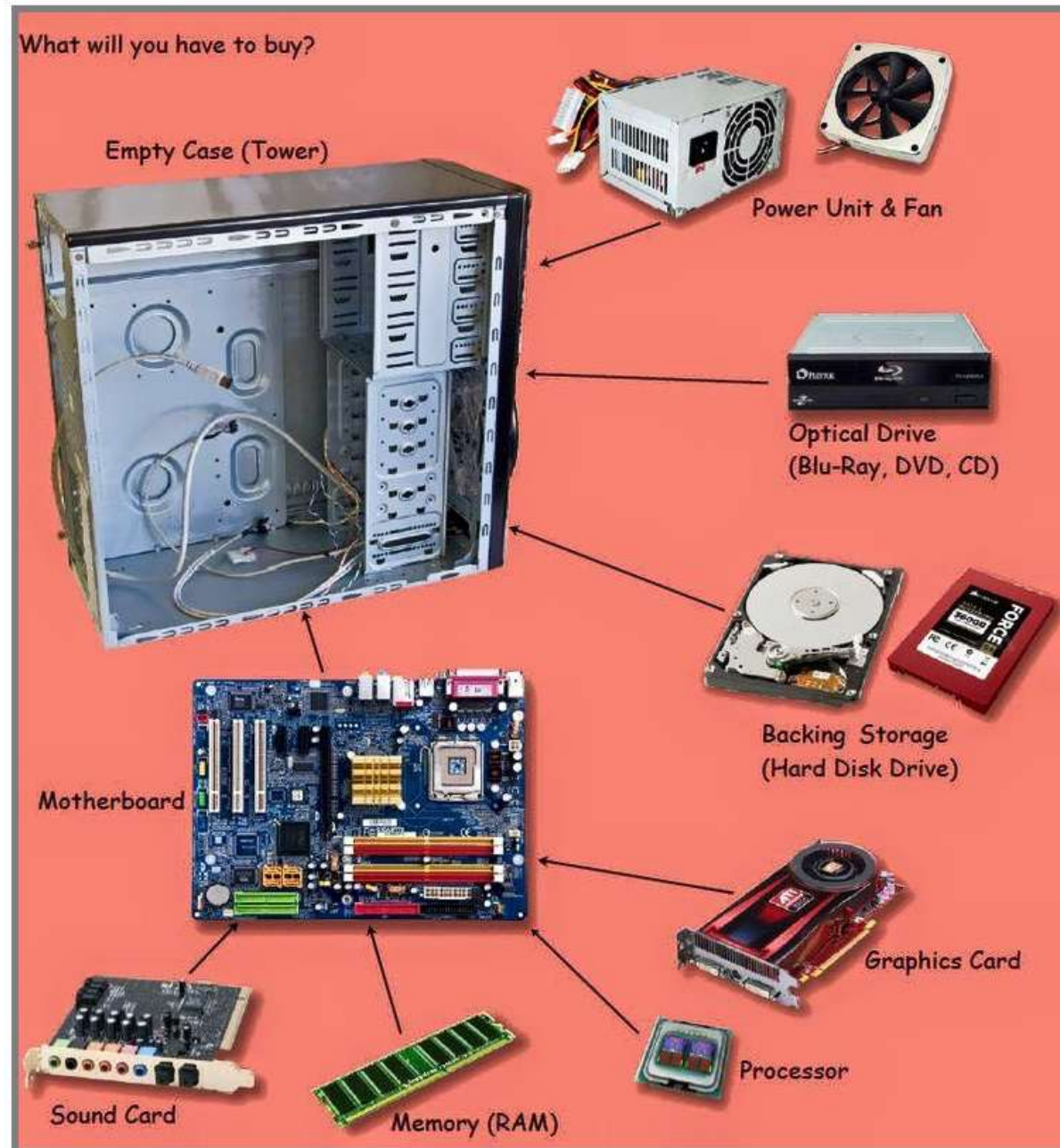
Micro USB B



USB Type C



Internal Components



Component Functions

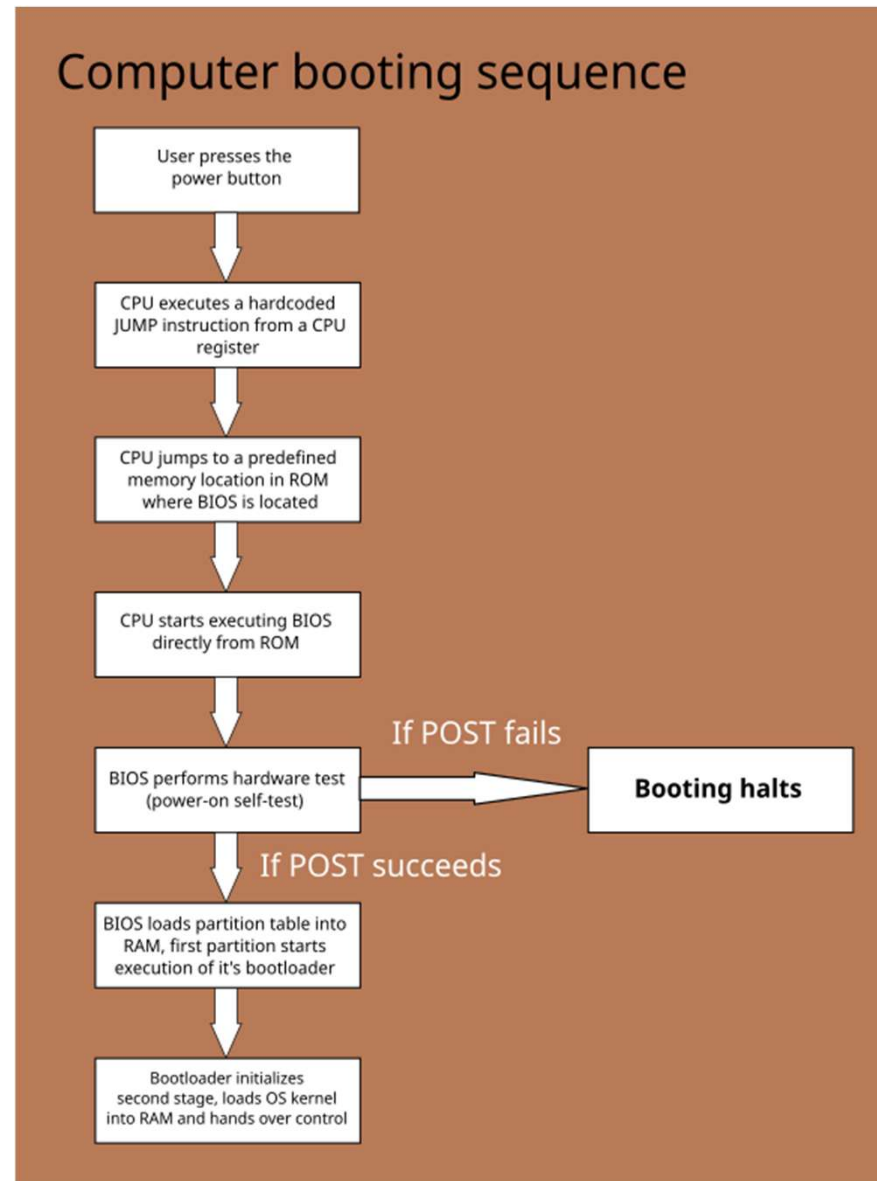
Component	Description	Function
Motherboard	Main circuit board	Connects all internal components and allows communication between them
Central Processing Unit (CPU)	The "brain" of the computer	Performs calculations and executes instructions
Random Access Memory (RAM)	Temporary memory	Stores data and instructions currently in use for fast access
Hard Disk Drive (HDD) / Solid State Drive (SSD)	Long-term storage	Stores operating system, applications, and files
Power Supply Unit (PSU)	Converts electrical power	Supplies power to all internal components
Monitor	Visual output device	Displays text, images, and videos generated by the computer
Keyboard	Input device with keys	Used to enter text and commands

Component Functions

Component	Description	Function
Mouse	Pointing device	Used to interact with graphical elements on screen
Graphics Processing Unit (GPU)	Optional separate chip or card	Renders images, videos, and animations (useful for gaming and design)
Optical Drive (CD/DVD)	Reads/writes CDs or DVDs (less common now)	Allows reading and writing of discs
Cooling Fan / Heat Sink	Temperature control components	Keeps CPU and other parts cool to prevent overheating
Speakers	Audio output devices	Play sound output from the computer
Network Interface Card (NIC)	Built-in or external card	Enables connection to a network (LAN/internet)
USB Ports	Universal connectors	Allow connection of external devices like pen drives, printers, etc.

Working of a Desktop Computer

Booting Process Overview

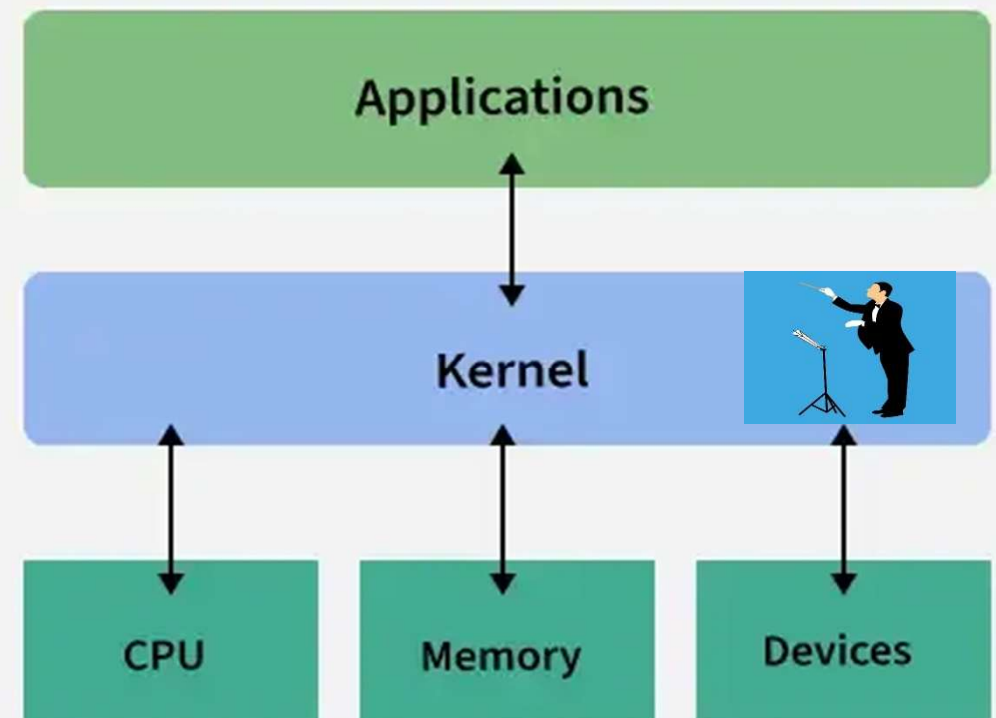


What Is an Operating System?

- An **Operating System (OS)** is the **main software** that manages all the **hardware** and **software** in a computer. It acts like a bridge between the user and the computer hardware.

Examples:

- Windows (most desktops)
- macOS (Apple computers)
- Linux (Ubuntu, Fedora, etc.)
- Android/iOS (in smartphones)



What Is an Operating System?

Function	What it Does	Example
Process Management	Runs and controls programs	You open a browser + Word
Memory Management	Allocates and tracks RAM use	Keeps apps from overwriting each other
File System Management	Organizes data into files and folders	Access your saved documents
Device Management	Controls hardware like printers, mouse, keyboard	Plugging in a USB drive
User Interface (UI)	Allows user to interact with system (GUI/CLI)	Windows desktop, Linux terminal

File System and Drives

- A **file system** is a method used by the Operating System to **store, organize, manage, and access files** on a **storage device** (like hard disks, SSDs etc.).

It defines:

- How data is **named**
- How it is **stored and retrieved**
- How files are **organized** (like folders and directories)

File System and Drives

Term	Explanation
File	A collection of data stored under a name (e.g., Word doc, image, video)
Folder (Directory)	A container that holds multiple files or folders
Path	The address of a file/folder (e.g., C:\Users\Yogi\Documents\notes.docx)
Drive	A letter assigned to a storage device (like C:, D:, E:)

Useful System Tools in a Desktop Computer

Tool Name	Purpose
Task Manager	View and manage running apps and processes; monitor CPU, RAM usage
File Explorer / Finder	Browse, organize, copy/move/delete files and folders
Control Panel / Settings	Change system settings: network, display, sound, users, updates
Command Prompt / Terminal	Run text-based commands for advanced operations

Basic Operations & Maintenance

Basic Desktop Operations

- Create/delete folders
- Organizing folders Demo
- Use keyboard shortcuts (Ctrl+C, Ctrl+V, etc.)
- Check system info (RAM, processor)
- Running setup (.exe) - Install options and safety tips - Uninstalling via Control Panel
- Restarting frozen systems - Antivirus and updates - Cleaning keyboard/screen and safe shutdown

Thank You!